

Unit 10, Lesson 5: Solving Two-Step Equations Using Properties of Equality

Benchmark

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.

Learning Targets

- ☐ I can relate properties of equality to solving two-step equations.
- ☐ I can solve two-step equations in one variable.

10.5.1 Warm-Up: Notice and Wonder

A math teacher and her daughter experimented with a seesaw at her local park. Their goal was to determine how the daughter could play on the seesaw on her own. She placed a crate on the other side of the seesaw and began adding bricks to it one at a time. Each time she added a brick, she tried to seesaw with the crate. A series of screenshots of her experiment are shown.

Seesaw with 0 Bricks



Seesaw after adding 1 Brick and a Crate

Brick - 1



Seesaw after adding 12 Bricks and a Crate

Bricks - 12





1. Complete the table to indicate one thing you notice and one thing you wonder when looking at this situation.

One thing I notice is...	One thing I wonder is...

2. Explain what is happening in this situation using the words “equal,” “greater than,” and/or “less than.”

10.5.2 Exploration: Properties of Equality

1. The equation $-6 + 10 = 4$ is used in the table to show different operations.
- a. Work with your partner to complete the table.

Operation	Step	Show Step	Resulting Equation
+	Add 5 to both sides of the equal sign	$-6 + 10 + 5 = 4 + 5$	___ = ___
-	Subtract 8 from both sides of the equal sign		
×	Multiply the terms on both sides of the equal sign by $\frac{1}{2}$		
÷	Divide the terms on both sides of the equal sign by 2		

- b. What do you notice about each of the resulting equations?
- c. With your partner, compare the results of multiplying each term in the original equation, $-6 + 10 = 4$, by $\frac{1}{2}$ and dividing each term by 2. Summarize your observations.

2. Complete the statement.

Dividing by 4 is the same as multiplying by ____ .



3. Use the equation $5\frac{1}{2} + 8 = 13.5$ to complete the following.

a. Discuss with your partner why the equation could also be written $13.5 = 5\frac{1}{2} + 8$. Summarize your discussion.

b. Work with your partner to complete the table for the equation $5\frac{1}{2} + 8 = 13.5$.

Operation	Step	Show Step	Resulting Equation
+	Add 23 to the left side of the equation and 14 to the right side		
–	Subtract 5 from the left of the equation and 10 from the right side		
×	Multiply the terms on the left side of the equation by 3 and those on the right by –1		
÷	Divide the left side of the equal sign by 0 and the right side of the equal sign by 0.5		

c. Prisha said it is not possible to complete the last row of the table. Explain why Prisha said it is not possible.

- d. What do you notice about the resulting equations in the first three rows of the table?

4. Complete the statements.

An equation compares two equivalent expressions using the mathematical symbol

= to show the values of both quantities are

- ☐ equal.
- ☐ unequal.

When the same operation is performed on both sides of an equation, the quantities

- ☐ remain
- ☐ are no longer

equivalent. When the operations

performed on both sides of an equation are not the same, the resulting equation is

true because the quantities being compared are

- ☐ still
- ☐ no longer

equivalent.

10.5.3 Bring It Together: Properties of Equality

Properties of equality can be used to solve equations.

1. Complete the table with a numeric example of each property of equality.

Property	Algebraic Statement	Numeric Example
Symmetric Property of Equality	If $a = b$, then $b = a$.	
Addition Property of Equality	If $a = b$, then $a + c = b + c$.	
Subtraction Property of Equality	If $a = b$, then $a - c = b - c$.	
Multiplication Property of Equality	If $a = b$, then $a \cdot c = b \cdot c$.	
Division Property of Equality	If $a = b$ and $c \neq 0$, then $a \div c = b \div c$.	

2. Explain why $c \neq 0$ for the division property of equality.
3. Which property justifies the equation $2m - 7.3 = 24.45$ is equivalent to $24.45 = 2m - 7.3$?

10.5.4 Guided Instruction: Solving Two-Step Equations Using Properties of Equality

When solving equations, the goal is to isolate the variable to find the value(s) of the variable that make the equation true.

The operations being performed on the variable can be “undone” using inverse operations and opposites in order to isolate the variable. Properties of equality can be applied to create equivalent equations using steps to isolate the variable.

1. Consider the equation $5x - 9.6 = 21.4$.

a. Complete the statement to describe the equation.

The variable, x , is first being _____ by 5, and then 9.6 is being _____ from this product. The value of the expression, $5x - 9.6$, is _____ to 21.4.

b. Describe the operation that could be performed to isolate $5x$ on the left side of the equation.

c. Show this step to create an equivalent equation.

$$5x - 9.6 + \underline{\hspace{1cm}} = 21.4 + \underline{\hspace{1cm}}$$

$$\underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

d. Describe the operation that could be performed to isolate the x on the left side of the equation.

e. Show this step to create an equivalent equation.

$$5x \div \underline{\hspace{1cm}} = 31 \div \underline{\hspace{1cm}}$$

$$x = \underline{\hspace{1cm}}$$

- f. What is the coefficient of the x after dividing both sides?
- g. Use substitution in the original equation to verify the solution.
 $5x - 9.6 = 21.4$
- h. Which properties of equality were applied to solve this equation?
- i. Cameron said the equation $5x = 31$ could have been solved by applying the multiplication property of equality. What value can be used as the factor that is multiplied to both sides of the equation to isolate x ?

2. Solve $-6.7 = \frac{y}{-4} - \frac{1}{2}$ by applying properties of equality.

- a. Fill in each box with an operation and each blank with a value.

$$-6.7 = \frac{y}{-4} - \frac{1}{2}$$

$$-6.7 \boxed{} \frac{y}{-4} - \frac{1}{2} \boxed{} $$
$$ \phantom{\boxed{}} \frac{y}{-4}$$

$$ \boxed{} \phantom{\frac{y}{-4}}$$

$$ \phantom{\boxed{}} y$$

- b. Use substitution in the original equation to verify the solution.

$$-6.7 = \frac{y}{-4} - \frac{1}{2}$$

- c. Which properties of equality were applied to solve this equation?

10.5.5 Try It: Solving Two-Step Equations Using Properties of Equality

Work with your partner to complete the following.

1. Use the equation $4.76 = -5 + 3.2n$ to complete the following.
 - a. Apply properties of equality to solve the equation.
 - b. Use substitution to verify the solution.

2. Use the equation $\frac{5}{7}r + 6\frac{3}{7} = -8$ to complete the following.
 - a. Apply properties of equality to solve the equation.
 - b. Use substitution to verify the solution.

10.5.6 Wrap-Up: Reflection

Complete the statements as you reflect on today's lesson.

1. Today I am still unsure about _____
_____.
2. To fill in this gap I intend to _____
_____.

